Selim Amar

Contact Information

Stanford, California, USA `selama@stanford.edu`

Summary

I am a mathematician with a background in analysis, partial differential equations, and physics.

Relevant Skills

Languages: English, French
Programming: Python — NumPy, SciPy, Matplotlib
Mathematics: PDEs, Microlocal analysis, Wave equations, Finite element methods

Education

Stanford University

PhD in Mathematics, (2022-Present); GPA: 4.0
Expected date of graduation: 2027

McGill University

B.Sc. in Honours Mathematics and Physics, (2019-2022); GPA: 4.0

Research Experience

2022-Present — Doctoral Research on Microlocal Analysis.
PhD Advisor: András Vasy, Department of Mathematics, Stanford University

Developed new analytic methods for non-elliptic PDEs. Proved Fredholm estimates for wave-type operators in the presence of normally hyperbolic trapping. Defined Feynman propagators and computed the index of Lorentzian Dirac operators. Techniques involve spectral theory, functional analysis, pseudodifferential operators and dynamics.

2021 — Peeling theorem in higher dimensions.
Supervisor: N. Kamran, Department of Mathematics, McGill University

Derived asymptotics for waves on higher-dimensional spacetimes using the Newman-Penrose formalism. Techniques involved asymptotic analysis, ODE theory and differential geometry.

2020 — Discrete exterior calculus.
Supervisor: G. Tsogtgerel, Department of Mathematics, McGill University.

Conducted numerical experiments on simulations of PDEs using the discrete exterior calculus and compared the observed convergence rate with the expected one. Techniques involved finite element methods, discretization of manifolds and numerical linear algebra. Coding done with Python.

Publications

S. Amar, *Feynman Propagators on Exponentially Decaying Perturbations of Stationary Spacetimes with Compact Cauchy Surface*
S. Amar, *Fredholm Theory of Non-Elliptic Operators in the Presence of Normally Hyperbolic Trapping*
S. Amar, *A Peeling Theorem for the Weyl Tensor in Higher Dimensions*, Classical and Quantum Gravity (2022).

Relevant Talks

Fredholm Theory of Non-Elliptic Operators with Trapping (McGill University Geometric Analysis Seminar and Berkeley Spectral Theory Seminar)
Neural networks and PDEs: from physics-informed learning to unstable solutions (Student Learning Seminar on Mathematics and Computation — Stanford University)

Selected Honors and Awards

FRQNT Doctoral Training Scholarship (2024–); Gowrisankaran Prize in Mathematics & Statistics (2022); Horace Watson Prize in Physics (2022); NSERC Undergraduate Research Award (2021).